Tremor Control Spoon (Steady Spoon)

Electronic design - excerpt and attachments

Biodesign course 2023-2024, Universita Campus Bio-Medico di Roma (Team 10).
Source: Final Report v4.0 (11/01/2024), pages 6-9 (section 2.5 Electronic Design), plus the PCB Design and Charge Circuit attachments.

Author of the electronic design: Marco Gallinaro.
The rest of the project (mechanics, data analysis, firmware, cost analysis) was developed by the team and is only summarised on the project web page.

surface there are: the USB port, the charging port and the chamfers to allow easy grip. The handle is connected to the front case with three M3 screws.
The case (Figure 5.b) allows the passage of the wires of the motors through a small window. Under the window, there is the motor dedicated to the compensation of roll, which is directly connected with a M1 screw to the case that holds the motor that compensates pitch. This second motor is connected to a spherical cap placed at 0.2 mm from an outer one; this system acts as a spherical joint. The maximum diameter of this component is 66 mm. Furthermore, on the outer external cap there is a level surface to allow the spoon to be easily placed on a table.
The tool (Figure 5.c) is joined to the rest of the spoon and can be easily removed for washing; the tolerance for the joint is 0.1 mm. It is placed on the same axis of the roll motor. It has a length of 70 mm and a maximum width of 32 mm.
To make the spoon waterproof, there is an elastic PP film that covers the spherical joint and fits inside the tool joint and the case.

[See CAD Design Attachment]

2.4.2. Materials and Manufacturing

The spoon is made of polypropylene, a polymer used to make kids' cutlery, which is therefore washable in the dishwasher, biocompatible, light, recyclable, heat resistant and has a high resistance to fatigue. All components but the film and the silicon caps are made of polypropylene high impact, the film is made of generic polypropylene and the caps are made of silicon.
The weight of the spoon is 81 g, which reaches a total of 133.4 g considering the servos and the battery.
The removable ports on the handle for USB communication and charging are in silicon.
The production of each piece is intended for injection molding; the parts in silicon are printed at room temperature.

[See Materials and Manufacturing Attachment]

2.5. Electronic Design

To miniaturize as much as possible the electronics of the system, we decided to integrate the three main components (sensing, control and power source) into one PCB. The electronic module was designed using Eagle (the schematic and the board can be found in the shared folder).
On the PCB, from left to right, there are:

- The battery pins (V_BATT and GND);
- A switch to choose between the charging mode (V_CHARGE) and the normal use mode (V_IN). The charging mode allows also the system to switch off when the charger is disconnected;
- A recharging circuit;

- A 5 V regulator;
- A 9 V charging jack connected to the charging circuit;
- A USB to serial UART interface (FT232RL) to load the firmware and supply 3.3 V regulation;
- The ATMega328-MUR chip as microcontroller of the whole platform;
- A USB-B 2.0 connector;
- The MPU-6050 sensor;
- The 6 pins for the two servos: for each motor, there is a digital pin used for the PWM signal, a 5 V power pin and a GND pin.

The board was designed starting from an Arduino Nano, deleting unnecessary components and adding custom ones.
We managed to reduce the original dimensions of the stripboard used in the demo (65 x 30 mm) to 57 x 31 mm, with an area reduction of 10%.

[See PCB Design Attachment]

2.5.1. Sensing Module

To monitor the hand tremor, a magnetic inertial sensor, capable of measuring accelerations and angular velocities, is needed on the board of the device. We chose the IMU MPU-6050 sensor, because it is the world's first integrated 6-axis Motion Tracking device which combines a 3-axis gyroscope, 3-axis accelerometer, in a small 4 x 4 x 0.9 mm package. It is easily available and largely used in literature.
The information is taken to the ATMega through the SDA and SCL lines with I2C protocol; these lines are connected to the 3.3 V with two 4.7 kΩ pull-up resistors. The 3.3 V power supply is connected with a bypass capacitor to attenuate possible noise.

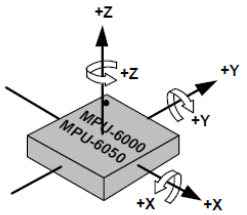


Figure 6. MPU-6050 sensor

2.5.2. Control Module

The ATMega328-MUR microcontroller is the main component of the control module, where the control code is loaded. The communication with the IMU sensor takes place through SDA and SCL lines and the PWM signal to the motors are taken from the digital pins PB4 (D12) and PB0 (D8).
There is also the external oscillator, the reset button and the 5 V power supply with bypass capacitors, necessary to attenuate the noise in the power signal.

2.5.3. Actuation Module

The two motors that compensate roll and pitch were chosen to have enough torque and angular velocity to effectively support and rotate the spoon tool during the meal. Their specifications were made using the dominant frequency, found with the data analysis.
First, we assumed the motors to move with a trapezoidal velocity profile; we then calculated the maximum angular velocity and the torques needed by each motor to move the spoon; to achieve this, we made assumptions about the shape of the spoon. We then considered the maximum torque and calculated the mechanical and electrical power needed by the load.
We chose the servos HD-1370A for their size. Furthermore, servos are easily controlled in position with a good accuracy without the need of adding a motor reduction and they are also cheap.

[See Motors Attachment]

2.5.4. Power Source

To supply the system, we chose to use a 2-cell 2S1P LiPo battery with a capacity of 800 mAh and a voltage of 7.4 V.
We considered two time features:

- The discharging time, approximately 2.36 h and calculated taking into account the current consumption of the whole PCB;
- The charging time, equal to 2 h and set by dimensioning the charging circuit.

The battery can be connected to the home network through the charging jack, compatible with the standard 9 V chargers.
For good power management, we inserted a voltage regulator. This component has two main tasks: voltage regulation according to load and voltage stabilization. The choice fell on the LM1117-5V, which supplies 5 V to the motors, the ATMega328 and the FT232RL integrated circuit, with a total output current of 324 mA.
A 3.3 V regulator is needed to power the sensor; the 3.3 V regulation is made by the FT232RL itself, since the MPU-6050 draws a very low current (3.9 mA), which the ICD is capable of supplying (up to 50 mA).

[See Battery Attachment, Charge Circuit Attachment and Regulators Attachment]

2.5.5. USB Communication

The USB allows to write the code in the memory of the ATMega328 microcontroller through the USB connector.
The connector is linked with the USB to UART interface, which allows communication thanks to its pins TX and RX that are connected to the microcontroller. The FT232RL is powered at 5 V from the regulator when the

spoon is used in normal mode and is powered at 5 V from the USB when the device is connected: in this case, a 5 V auto selector brings the USB voltage into the chip, stabilizing the voltage.
Moreover, the FT232RL is used to regulate the 3.3 V so that the IMU sensor can be powered at that voltage.

2.6. Firmware Description

The firmware follows the control logic in Figure 7: the IMU sensor collects acceleration and angular velocity; in the planning module, the complementary filter is implemented to obtain the roll and pitch angles (ϕ and θ); in the act module, the motors are moved to compensate roll and pitch.

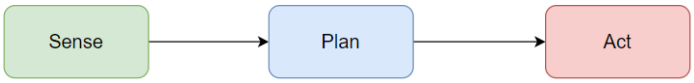


Figure 7. Firmware control flow

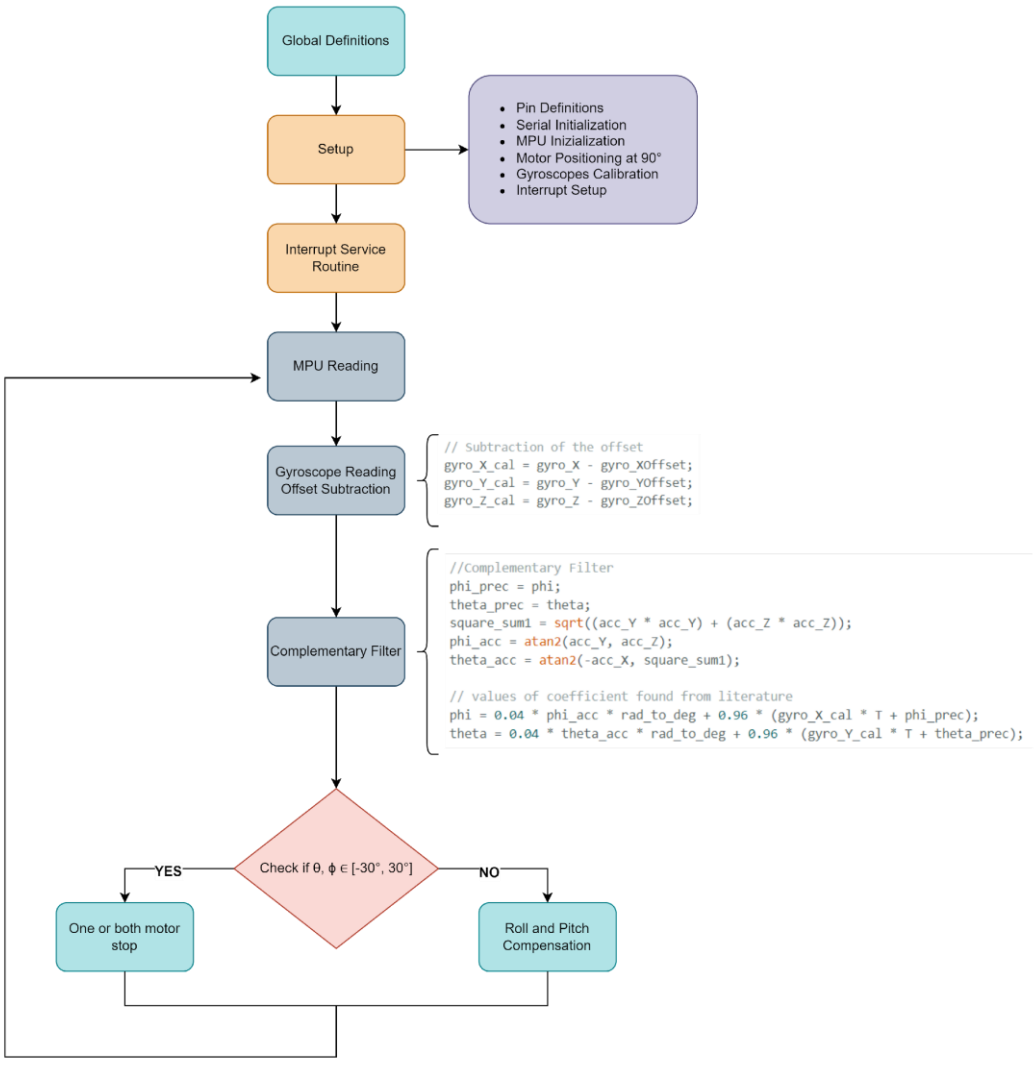
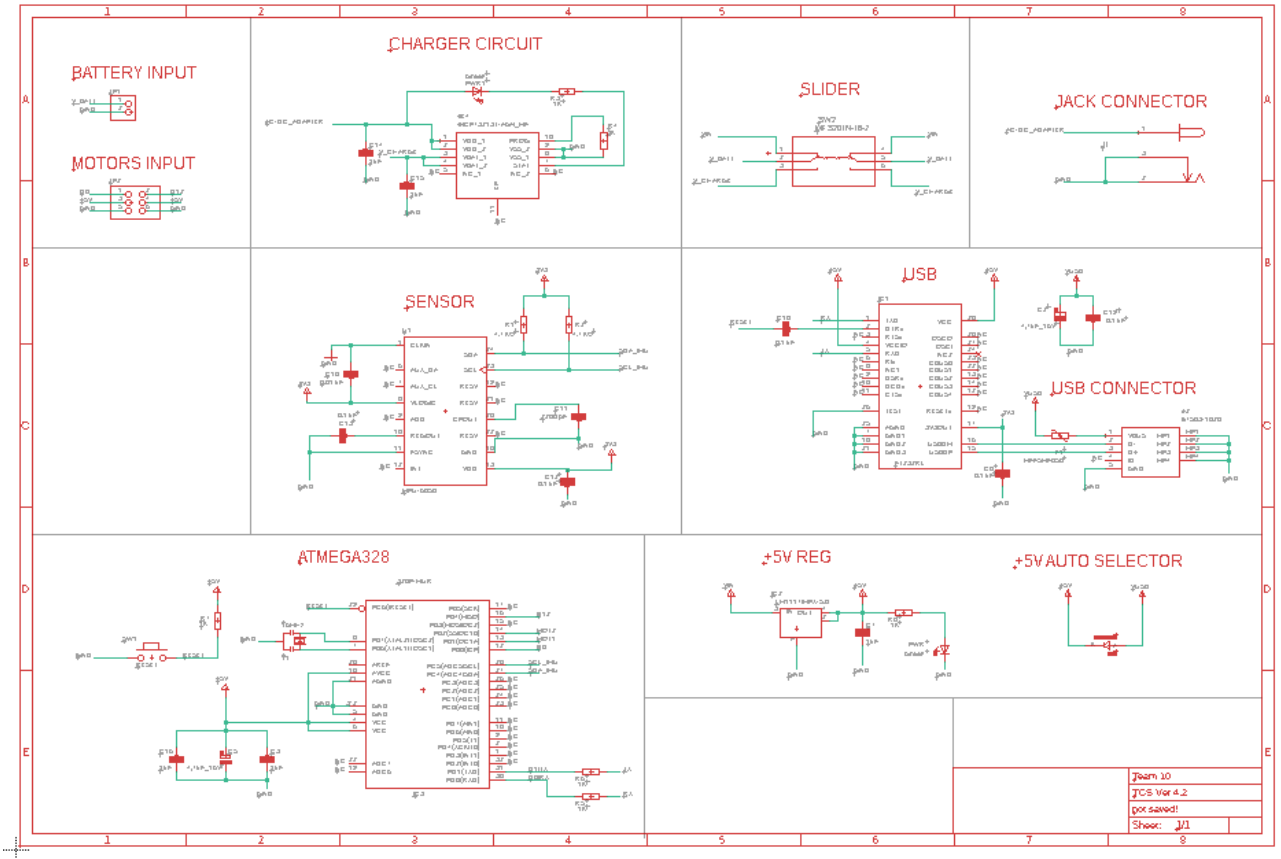
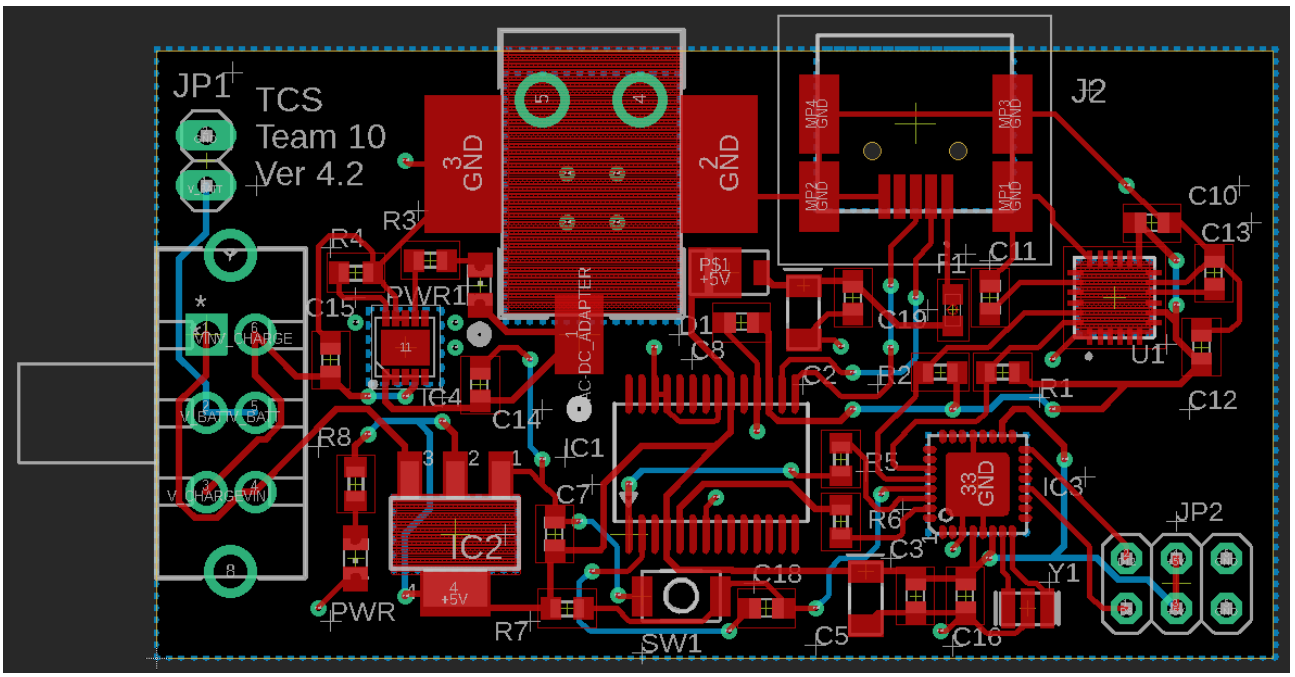


Figure 8. Firmware block diagram

PCB Design





The PCB was designed as the figure above shows.
On the left side, there is a switch that can be used to switch between the recharging mode and the normal use mode.

Battery Pins

The battery is connected through a JST Plug with two pins.

Charge Circuit

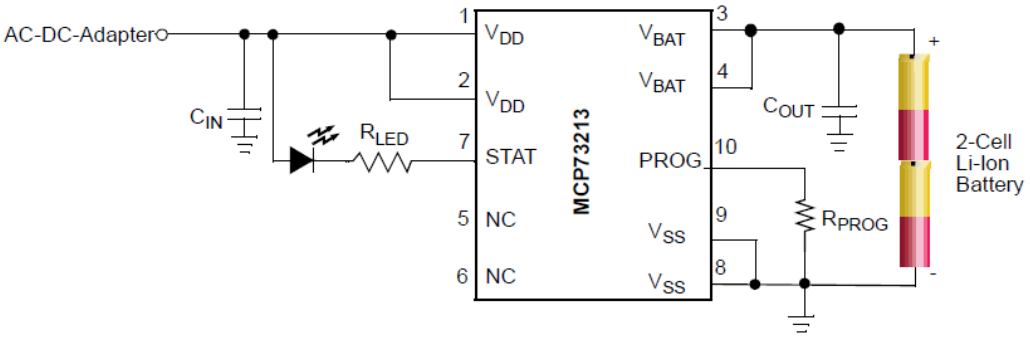
The charge circuit is made of the MCP73213, two 1 µF capacitors to discharge voltage oscillations to ground, a 3 kΩ resistor to limit the current given to the battery [see Charge Circuit Attachment] and a led with a resistor to show that the battery is recharging. The resistance in series to the led was chosen taking into account

$$R_{LED} = \frac{V_{CHARGE} - V_{LED}}{I_{LED}}$$

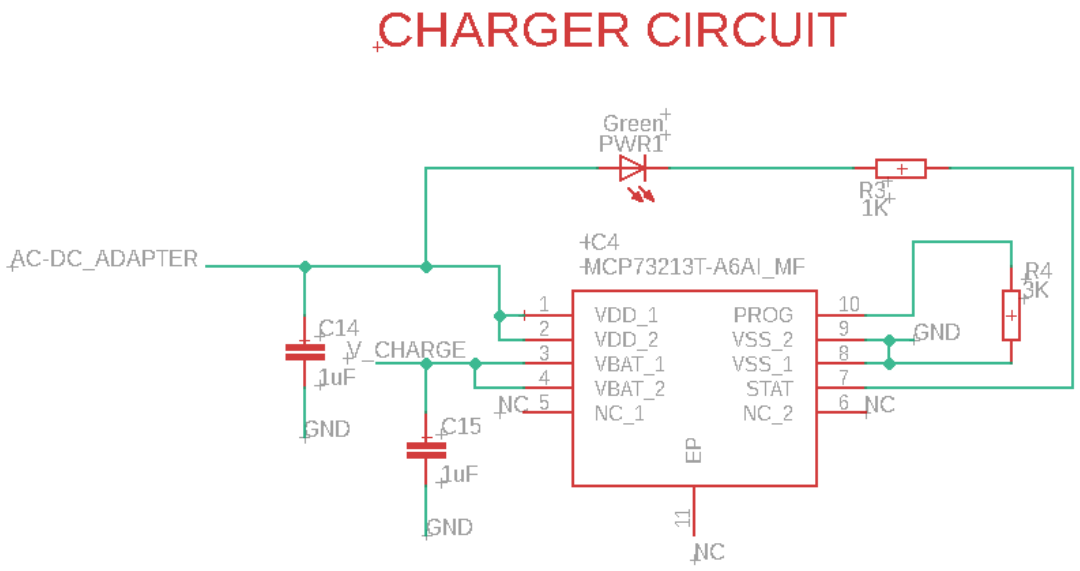
where V_{CHARGE} is the tension (9 V) applied to the positive of the led when the charger is connected, V_{LED} is the forward voltage of the led (2.2 V from the datasheet) and I_{LED} is the typical current on the led (20 mA). Therefore

$$R_{LED} = \frac{9\ V - 2.2\ V}{20\ mA} = 340\ \Omega$$

This resistor must limit the current, so it is acceptable to choose one with a higher value (1 kΩ). As per the MCP73213 design guidelines, the capacitors were put as close as possible and there are some vias to allow heat dispersion.



Typical Application (from the datasheet of MCP73213)

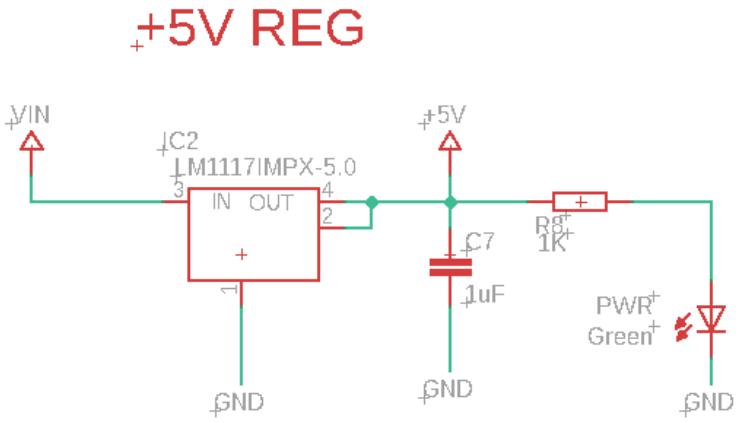


Charging Jack

The charging jack also has heating issues, so there are no traces under it. A keep-out zone has been placed on the space it occupies and some vias are used for heat dispersion as well.

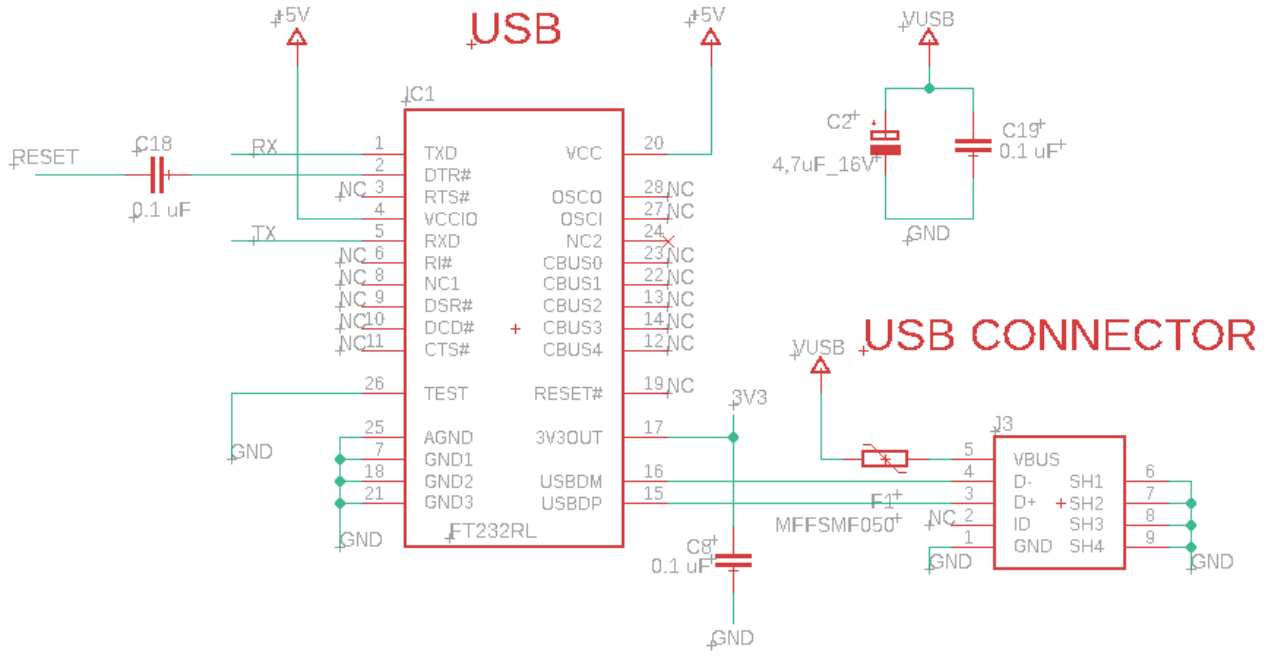
5V Regulator

The 5 V regulator LM1117 is placed with a bypass capacity with a value of 1 µF, as advised by the datasheet. A led was added with a 1 kΩ resistor to limit its current, in the same way as above. Traces carrying the load current are wide to reduce the amount of parasitic trace inductance and the feedback loop from pin 4 to 2 is as short as possible. There are also vias for heat dispersion. There are no traces on the top layer under the integrated circuit.



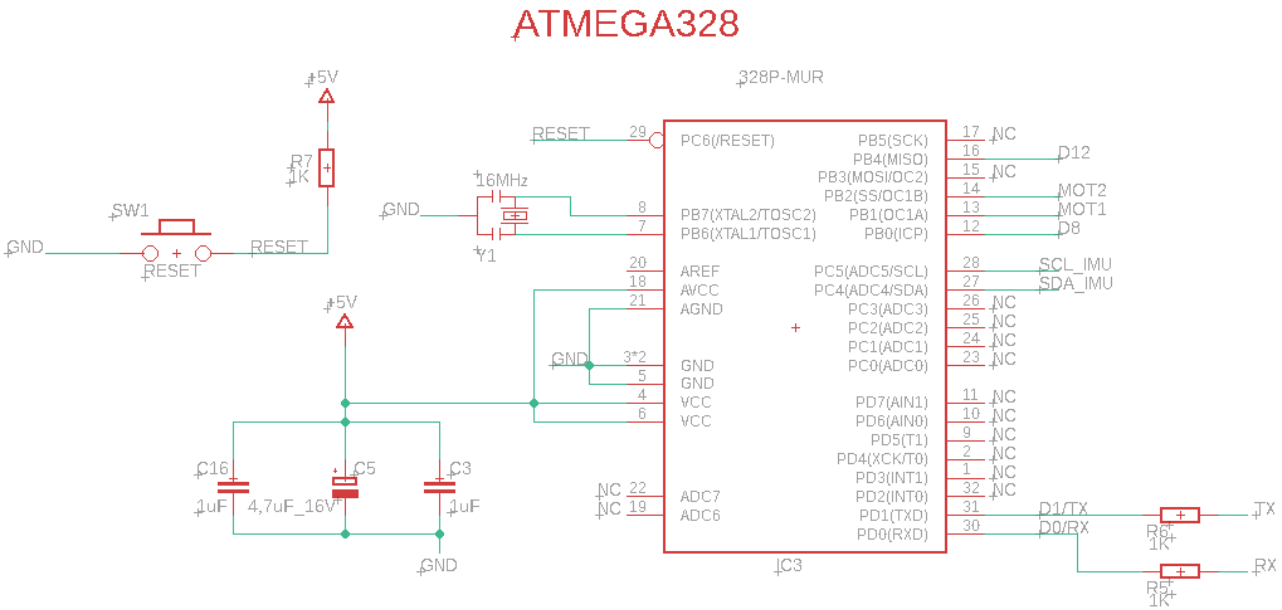
USB

The USB module is made of the USB-B connector and the FT232RL integrated circuit. The bypass capacitors C2 e C19 are placed close to VUSB, while C8 and C18 are close to the FT232RL. C8 is a bypass capacity on the 3.3 V line and prevents noise; C18 is linked with the reset button and limits the debouncing effect by filtering. The 3V3 trace is wider than the others.
VUSB is brought to the connector with a PTC (positive temperature coefficient) fuse that protects the PCB from high temperatures.



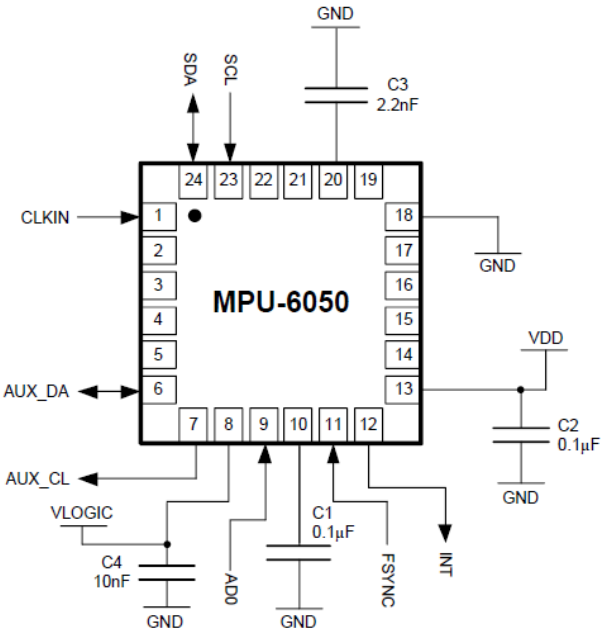
Microcontroller

The ATMega328 is placed as close as possible to the bypass capacitors C3, C5 and C16, and the external oscillator is close as well. R6 and R7 are, respectively, pull-up (on the transfering line) and pull-down (on the receiving line) resistors that make sure the communication between the processor and the USB serial is successful. Communication with the IMU sensor is made possible by the SDA and SCL lines; to the RESET pin is connected a reset button, which is open when it is not pressed and closed otherwise (NO, normally open).

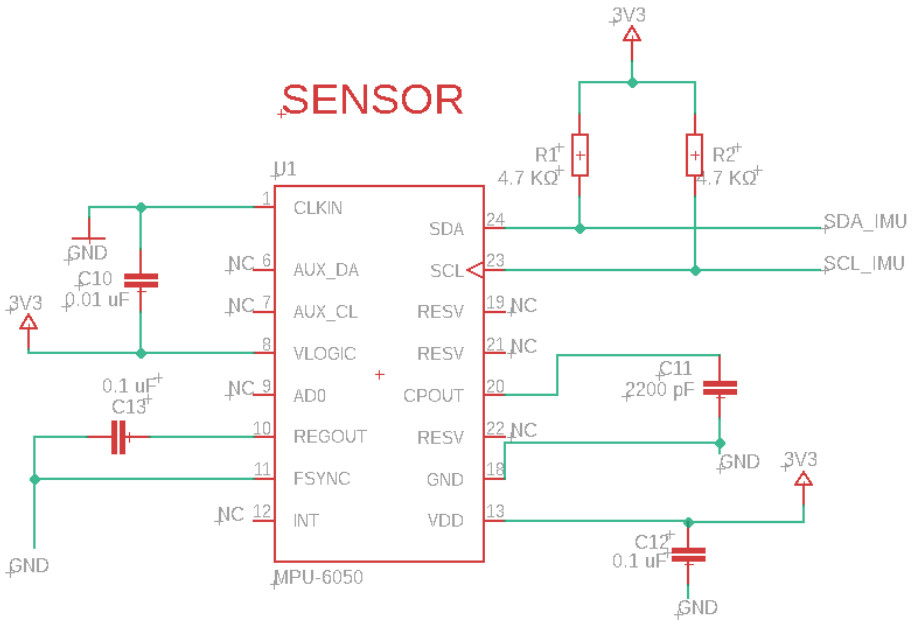


IMU Sensor

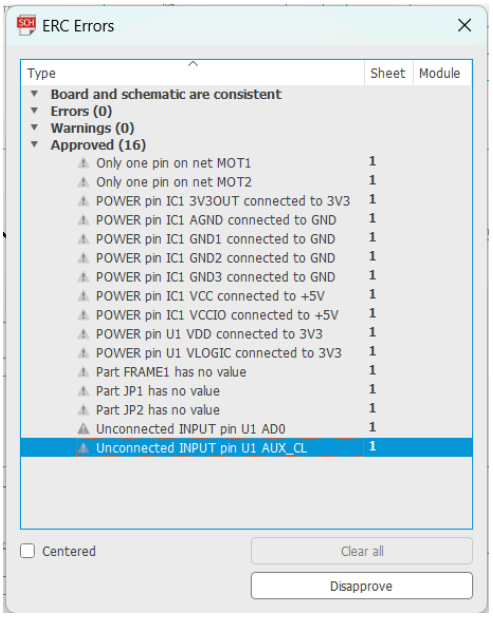
The MPU-6050 sensor is placed far from the reset button, as suggested by the guidelines in the datasheet. The bypass capacitors C10, C11, C12 and C13 are placed as close as possible.



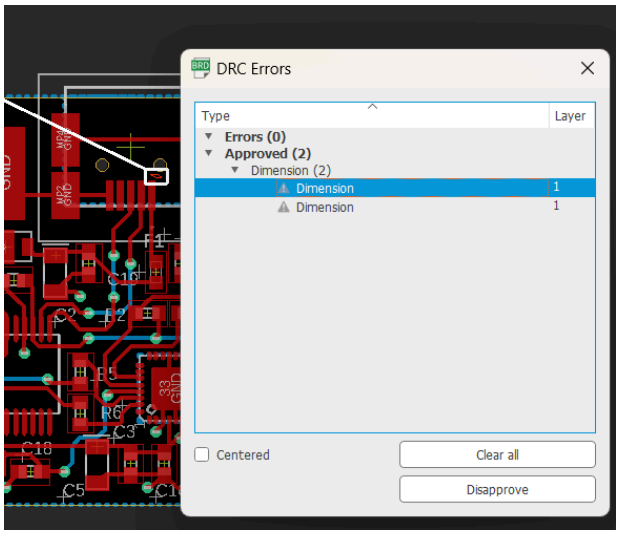
Typical operating circuit (from datasheet)



On the schematic, there are 16 approved ERC warnings.



On the board, there are 2 approved DRC errors: these are caused by the design of the footprint of the USB connector, so they cannot be solved.



Charge circuit

The charging cycle of the battery is generally divided into two phases: a constant current (CC) phase and a constant voltage (CV) phase.
If the voltage of the battery is too low, there is also the CC pre-conditioning phase, in which a small current is given to the battery until the voltage exceeds a threshold value. In this condition, the current must be low to limit the Joule effect. After the pre-conditioning phase, there is the CC phase until the battery voltage reaches the 95% of the nominal battery voltage. Finally, there is the switch to the CV phase, that lasts until the current reaches zero.

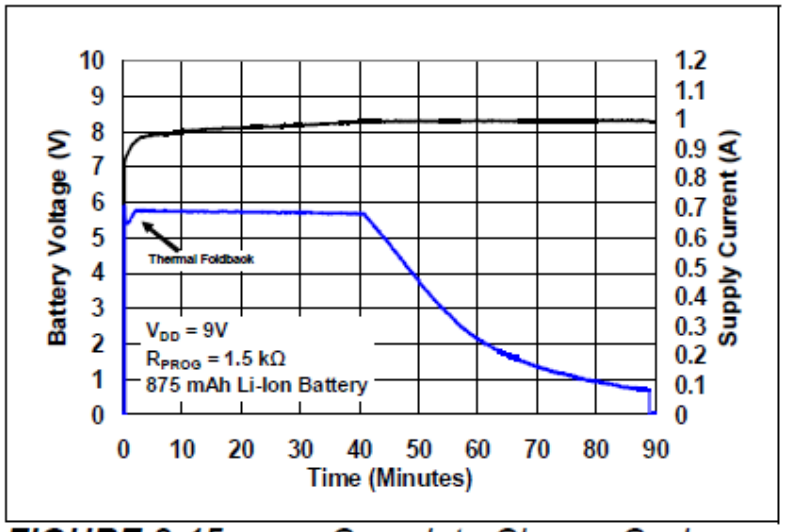


FIGURE 2-15: *Complete Charge Cycle.*

The figure above is taken from the datasheet of the MCP73213 charging circuit: the blue curve is the current, while the black curve is the voltage.

The battery we chose has a maximum charging current of 400 mA, while the maximum charging voltage has a standard value of 8.4 V. The chosen charging circuit must therefore fall within these ranges.

The MCP73213 is a charging circuit that has a charging voltage of 8.4 V and programmable charging current between 130 mA and 1100 mA. The current must then be adjusted so that it has a value of 400 mA. This setting is possible by choosing the value of the resistor R_{PROG} according to the formula below given in the datasheet:

$$R_{PROG} = 10^{\frac{log\left(\frac{I_{REG}}{1104}\right)}{-0.93}} = 10^{\frac{log\left(\frac{400\ mA}{1104}\right)}{-0.93}} = 2.98\ k\Omega$$

By choosing a resistor of 3 kΩ, a current value below 400 mA is achieved:

$$I_{REG} = 1104\ {R_{PROG}}^{-0.93} = 397\ mA$$

This means that the charging time of the battery is:

$$t_{CH} = \frac{C}{I_{REG}} = \frac{800\ mAh}{397\ mA} = 2\ h$$

From the datasheet, it can be seen that to obtain a charging voltage of 8.4 V, the input voltage to the circuit must be between 8.4 V + 0.3 V = 8.7 V and 13 V, so a 9 V connector can be used.